Scope and Requirements Specification

1. Scope

The software is intended to serve as a numerical simulation and experimentation tool for investigating the two-dimensional brusselator reaction-diffusion system. Its primary purpose is to help users understand the behaviour of coupled nonlinear reaction-diffusion equations through numerical modelling. By providing different numerical integration methods, configurable simulation parameters and automated parameter studies, the software enables users to investigate the influence of numerical schemes and physical parameters on the resulting concentration fields.

1.1 In Scope

- Numerical solution of the two-dimensional Brusselator reaction-diffusion system.
- Implementation and comparison of Explicit Euler, Runge-Kutta and Implicit Euler methods.
- Configuration of simulation parameters, including grid dimensions, time step and reaction parameters.
- Automated parameter studies to investigate the influence of selected simulation parameters.
- Generation of simulation results in VTK format for subsequent visualization and analysis using ParaView.
- Numerical validation against provided reference data in the resources directory using jupyter notebook launched from inside python virtual environment inside the docker container.

1.2 Out of Scope

- Solving arbitrary partial differential equations or reaction-diffusion models beyond the implemented Brusselator system.
- Performing three-dimensional simulations.
- Supporting arbitrary geometries and unstructured meshes.
- Providing an integrated graphical use interface for configuring and executing simulations. The solver is operated through command-line instructions. However, the generated VTK files can be visualized using the external application ParaView, which provides graphical tools for inspecting and analysing simulation results.
- Automatically identifying the optimal numerical integration method or guaranteeing numerical accuracy for arbitrary configurations.
- Providing unrestricted parameter studies beyond the variations implemented in the current software.

2. Functional Requirements (FR)

ID	Requirement
FR-01	The software shall solve the two-dimensional Brusselator reaction-diffusion system using numerical methods in a structured uniform mesh.
FR-02	The user shall be able to select between Explicit Euler, Runge-Kutta and Implicit Euler integration methods. These are the numerical methods deployed.
FR-03	The software shall accept user-defined grid dimensions, simulation duration, timestep, output write interval and reaction parameters Ca and Cb.
FR-04	The software shall validate simulation inputs and reject invalid configurations.
FR-05	The software shall generate simulation results in VTK format at specified output write intervals.
FR-06	The software shall support automated parameter studies to investigate time-step variations for the implicit solver and reaction-parameter Cb variations for the Runge-Kutta solver.
FR-07	The generated numerical results shall be capable of being compared against reference data for validation in jupyter notebook.
FR-08	The software shall generate VTK-compatible output files that can be opened in ParaView, enabling users to graphically visualize and analyse the spatial distribution and evolution of the simulated concentration fields.

3. Non-Functional Requirements (NFR)

ID	Category	Requirement
NFR-01	Accuracy and reliability	The software shall produce numerically consistent results, with validation against reference solution for Explicit Euler with an L2 error value of 3.2e-08
NFR-02	Performance	Numerical computations shall use memory-efficient data structures and appropriate numerical algorithms to handle sparse matrices efficiently. Eigen was used in this respect.
NFR-03	Portability	The software shall support a reproducible build and execution environment through clearly defined dependencies and a containerized Ubuntu environment. CMAKE, GNU MAKE and Docker was used in this respect
NFR-04	Maintainability and testability	The software shall follow a modular structure,allowing individual components to be modified and tested independently.
NFR-05	Usability	The software shall provide clearly documented execution instructions, simulation parameters, parameter study procedures and output organisation. Generated results shall be accessible through standard scientific visualization tools such as ParaView.